SAMEER POKHREL

PhD Candidate | Plant Breeding, Genetics and Genomics, University of Georgia
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Please click the link to: **Personal Website** **GitHub** **Google Scholar**

Professional summary

PhD researcher in plant breeding and genomics, working at the interface of peanut breeding, varietal development, trait discovery, pangenomics and facilitating open-source genomic research through bioinformatic pipelines

Academic qualification

2022 – current | PhD in Plant Breeding, Genetics and Genomics | UGA
Additional master's level certificate in Bioinformatics from Institute of Bioinformatics
Dissertation: Utilization of multi-parental peanut population to understand the genetics and develop drought tolerant and white mold resistant peanut lines
2019 – 2021 | Master's in Horticultural Sciences | University of Florida | GPA: 4.0/4.0
Thesis: Horticultural attributes and root architecture of field-grown Valencia citrus trees grafted on rootstocks propagated by seed, cuttings and tissue culture
2013 – 2017 | B.S. in Agriculture | Agriculture and Forestry University, Nepal **|** GPA: 3.6/4.0
Thesis: Economic analysis of maize seed and grain production in Rolpa district, Nepal

Work experience

2022 (Aug)- Current
Graduate Research Assistant | Ozias-Akin's lab, UGA-Tifton Campus
Advisor: Dr. Peggy Ozias-Akins

- Development of 18 parental peanut population and a population specific pangenome for high-resolution discovery of complex traits
- Developed an open-source interactive R shiny app that converts complex pangenomic data into user-friendly visualization of genetic variation and genotype clustering (link)
- Developed 3-5 promising drought-tolerant peanut lines over years through breeding; currently working on germplasm release through USDA
- Developed 5-7 white mold resistant peanut lines through field breeding with germplasm release targeted by the end of 2026
- Collaborated on the identification of a candidate gene associated with Tomato Spotted Wilt Virus resistance in peanuts and identification of locus associated with nematode resistance
- Collaborated on identification of two genes conferring resistance to root knot nematodes in peanuts.

2021 (Aug)- 2022 (June)
Graduate Researcher | World Food Crops Breeding Program, University of Florida
Advisor: Dr. Ali Babar

- Managed seed planting and machine harvesting, conducted detailed agronomic data collection, and controlled crossing for wheat breeding trials
- Maintained breeder's trial and trained in selection techniques for identifying superior genotypes in the trial.

2019 (Jan) – 2021 (July)
Graduate Research Assistant | Plant physiology lab, University of Florida, SWFREC
Advisor: Dr. Ute Albrecht

- Conducted multi-location and multi-year field evaluations of citrus root and shoot growth to assess rootstock performance in disease pressure
- Performed laboratory experiments including DNA extraction and qPCR to support physiological experiments.

2018 (Jan) – 2018 (Aug)
High school teacher | Shree Raghunath Adarsha Secondary School, Kailali, Nepal

- Instructor of the courses 'Agronomy' and 'Fruit crop production'
- Designed course materials and practical training guidelines

Research publications

Peer-Reviewed Publications

- Lee, K., Korani, W., Bentz, P. C., **Pokhrel, S**., Ozias-Akins, P., Harkess, A., et al. (2025). Long-read low-pass sequencing for high-resolution trait mapping. *G3 (Genes, Genomes, Genetics)*.
- Web, S., Chu, Y., Biswal, A., Korani, W., Botton, S., **Pokhrel, S**., Thompson, E., Guo, B., Culbreath, A., Clevenger, J., Holbrook, C., & Ozias-Akins, P. High copy number of an *Arachis hypogaea* gene is associated with increased resistance to Tomato spotted wilt virus. *Crop Science*, 66(3), e70306. https://doi.org/10.1002/csc2.70306
- **Pokhrel, S**., Kharel, P., Pandey, S., Botton, S., Nugraha, G. T., Holbrook, C., & Ozias-Akins, P. (2025). Understanding the impacts of drought on peanuts (*Arachis hypogaea* L.): Exploring physio-genetic mechanisms to develop drought-resilient peanut cultivars. *Frontiers in Genetics*, 15, 1492434. doi: 10.3389/fgene.2024.1492434
- **Pokhrel, S**., Pandey, S., Ghimire, A., & Kandel, S. (2021). Understanding citrus greening disease and its possible management strategies in Nepal. *International Journal of Applied Sciences and Biotechnology*, 9(4), 227–234. https://doi.org/10.3126/ijasbt.v9i4.40805
- **Pokhrel, S**., Meyering, B., Bowman, K. D., & Albrecht, U. (2020). Horticultural attributes and root architectures of field-grown 'Valencia' trees grafted on different rootstocks propagated by seed, cuttings, and tissue culture. *HortScience*, 55, 1–10. https://doi.org/10.21273/HORTSCI15507-20

Manuscript (accepted)

- Biswal, A. K., Quiroz, S., **Pokhrel, S.**, Flores, I. M., Gomez, L., Ozias-Akins, P., & Alakonya, A. Draft genome assembly of *Monographella maydis* predicts a molecular driver of sexual fitness in maize tar spot disease. *BMC genomics*. *In press*. Pre-print: https://assets-eu.researchsquare.com/files/rs-9543400/v1_covered_a0c461a1-0081-427d-83f1-f271ea5884c6.pdf?c=1779024634

Manuscripts (under submission)

- **Pokhrel et al.** Founder-resolved pangenomics identifies an NC94022-derived white mold resistance haplotype and allele-frequency shifts under phenotypic selection in peanut (*A. hypogaea*). The Plant Genome.
- **Vaughn, Pokhrel et al.** (shared first authors). Complete haplotype knowledge in a PanMAGIC peanut population enables Rapid Intercross Selection Engine (RISE).

Manuscripts (under preparation)

- **Pokhrel et al.** Genomic prediction of drought response and signature of selection in an eight-way MAGIC peanut population.
- **Pokhrel et al.** Registration of [xx], white mold-resistant peanut germplasm lines develop from an eight-way MAGIC population

Extension and media publications

- Root architecture, propagation method, and citrus growth, Citrus industry Magazine (link)
- Valencia Performance and Rootstock Propagation Methods, Citrus Industry Magazine (link)
- An incurable disease has been invading citrus orchards, The Kathmandu Post (link) Harmony and Discord, The Kathmandu Post (link)
- Climate change threatens 'Himalayan Viagra' fungus, and a way of life, The Guardian (link)
- Nepal needs to shift towards energy-efficient brick kilns, new spotlight magazine (link)

Honors and awards

- Selected as a mentee in ASA-CSSA-SSSA (tri-society) peer review mentorship program Spring, 2025 for training in scientific peer review, manuscript evaluation, and understanding publication ethics under the guidance of a mentor.
- Recipient of George Hugh Boyd Memorial Scholarship; 2023-2024, Graduate school, UGA. Renewed on a competitive basis for academic years 2024-2025 and 2025-2026.
- Travel awards to present at several scientific conferences, received from: PBGG; 2023, 2024, UGA Graduate School; 2024, American Society for Horticultural Sciences (ASHS); 2020, National Association of Plant Breeders (NAPB); 2025

- Full merit scholarship to study a Bachelor of Science in Agriculture, Agriculture and Forestry University, Nepal (2013-2017) | scholarship covered all the expenses related to admission and tuition throughout the undergraduate program

Research grants submission

- John Ingle Innovation Award in plant breeding, 2024/2025 (Awarded $5000)
  Project: Developing Drought-Tolerant and Aflatoxin-Resistant MAGIC Peanuts Using Drone-Assisted Selection
- Utilizing Multi-Parental Population and Sensor-Based Technologies to Develop Drought Resilient Peanuts with Potential Aflatoxin Resistance. Southern SARE 2025. (Not awarded)
- DNA sequencing and developing a pangenome graph of divergent cultivated peanut (*Arachis hypogaea*) populations. NIFA's AG2PI 2023. (Not awarded)

Research presentations

Delegate and invited speaker

Advances in Arachis Through Genomics and Biotechnology (AAGB), Goa, India. "Testing Agile Genetics in a complex multi-parental peanut population". Oral presentation. 2025.

Oral Presentations

Pokhrel, S. (2025). Testing Agile Genetics in a complex multi-parental peanut population. *Advances in Arachis Through Genomics and Biotechnology (AAGB)*, Goa, India. (Invited oral presentation)

Pokhrel, S. (2025). Breeding drought-tolerant peanuts to reduce aflatoxin contamination. *United States Peanut Federation (USPF)*. (Oral presentation)

Pokhrel, S. (2025). Leveraging MAGICally developed peanuts for drought resilience. *PBGG retreat*. 2025.

Pokhrel, S. (2020). Comparison of field performance of citrus trees on rootstocks propagated by seedling, cuttings, and tissue culture. *American Society for Horticultural Science (ASHS)*. (Oral presentation)

Pokhrel, S. (2020). Horticultural attributes and root architectures of field-grown 'Valencia' trees. *Florida State Horticultural Society (FSHS)*. (Oral presentation)

Pokhrel, S. (2018). Comparative socio-economic analysis of maize seed and grain production in Rolpa. *Y-PARD Nepal Research Symposium*. (Oral presentation)

Poster Presentations

Pokhrel, S. (2025). Testing the components of Agile Genetics in peanut PanMAGIC populations. *Plant and Animal Genome Conference (PAG)*.

Pokhrel, S. (2024). A Pan-MAGIC platform for peanut genetics study. *National Association of Plant Breeders (NAPB)*.

Pokhrel, S. (2024). Unleashing the power of pangenome in MAGICal peanuts. *PBGG Retreat, University of Georgia*.

Pokhrel, S. (2023). Genome assembly and comparative genomic analysis of tetraploid peanuts. *AAGB*.
Pokhrel, S. (2023). An efficient approach for tetraploid genome assembly. *PBGG Retreat, UGA*.
Pokhrel, S. (2021). Influence of rootstock propagation method and rootstock cultivar on citrus performance. *ASHS* (e-poster).
Pokhrel, S. (2019). Comparison of field performance of citrus trees on rootstocks. *South Florida Graduate Research Symposium*.

Professional membership

- Member of tri-societies (ASA, CSSA, SSSA) since 2025
- Member of National Association of Plant Breeders (NAPB) since 2024
- Member of PBGG, Graduate Student Association since 2022 and served as a secretary in 2023-2024 fiscal year
- Past member of American Society for Horticultural Science (2019-2022), Florida State Horticultural Society (2019-2021), Graduate Student Organization at SWFREC, UF (2019-2021), Gator citrus club (2019-2020)
- Past member and travel award chair of Plant Science Council, UF (2019-2021)

Service activities and leadership

Young Scholars Program mentor, UGA-Tifton campus | 2026
Mentoring a high school student in peanut genetics research to support hands-on scientific research and strengthen the learning environment at Tifton campus
Secretary, PBGG-Graduate Student Association, UGA | 2023-2024
Facilitated graduate student activities on Tifton campus, coordinated meetings, maintained records, disseminated information, and supported program planning
Core planning committee member, PBGG retreat hosted by Tifton Campus | 2023-2024
Helped plan and organize the annual retreat, supported graduate student and faculty engagement, assigned as a moderator for scientific talks of invited speakers at the annual retreat
Student events volunteer | UGA
Active volunteer in academic and student activities including seminars, journal clubs, speaker visits, campus programs and student-organized events.
Research paper review | 2025-2026
Through peer review mentorship program by ASA-CSSA-SSSA, receiving trainings on publication ethics, responsible contribution to the scientific peer-review process, and evaluating manuscripts for the societal journal. Independently peer-reviewed multiple manuscripts submitted to The Plant Genome journal after the end of the program.
Session moderator, ASHS | 2020
Moderated the citrus session during the ASHS annual scientific meeting
Travel award chair, Plant Science Council, UF | 2019-2020
Evaluated graduate student travel award applications, coordinated fair and timely award decisions, and supported plant science symposium

Post-earthquake agricultural recovery support, Nepal | 2016
Assisted farmers in restoring crop production systems in earthquake-affected region, Kaule, Chitwan, Nepal. Also advocated for resilient agricultural systems through writing and public engagement

References

Dr. Peggy Ozias-Akins
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